Probabilistic Artificial Intelligence
Final Exam

Jan 30, 2021

Time limit:	120 minutes
Number of pages:	19
Total points:	43

You can use the back of the pages if you run out of space. Collaboration on the exam is strictly forbidden. Please write your answers with a *pen*.

Rules and Instructions: Every correct answer gets points. There are no negative points for incorrect answers. Collaboration during the exam is strictly forbidden.

You will encounter the following question types.

- **Single choice questions.** There is only one correct answer out of several (typically, but not necessarily, 4). Said differently, no multiple choice question contains multiple true answers.
- **Groups of multiple True/False questions.** Every True/False question in a group of multiple True/False questions is evaluated separately. For example, if you answer 3 True/False correctly out of a group of 4, you get 75
- **Numerical answers.** Answer using the decimal point system (not fractions). You can round up or down to two decimal places CJ: (i.e. the tolerance is +/- 0.01)?. Only the final answer counts for the grade.

Notation. The notation $\mathcal{N}(\mu, \sigma^2)$ will designate a Gaussian random distribution with mean μ and variance σ^2. For example, $\mathcal{N}(3, 4)$ designates the Gaussian distribution with mean 3 and standard deviation 2.

Good luck!

(0 points) Please fill in your student ID and full name (LASTNAME, FIRSTNAME) in capital letters.

Please leave the table below empty.

Problem	**Maximum points**	**Obtained**
1.	18	
2.	12	
3.	8	
4.	9	
5.	15	
6.	9	
Total	43	

1. Bayesian Linear Regression (6 points)

The island Laputa had never seen rabbits until, one day, rabbits escaped one of the many touristic ships. Your know that the population of rabbits is given by

$$x_{t+1} = \omega x_t + \epsilon, \tag{M}$$

where x_t denotes the number of rabbit **pairs** in month t, ω is some parameter describing the birth rate, and ϵ is some additive noise.

(1 point) (i) Assume that the birth rate is $\omega = 2$, and that the noise in the model is i.i.d. and has distribution

$$\begin{aligned} P(\epsilon = -2) &= 0.4, \\ P(\epsilon = -1) &= 0.4, \\ P(\epsilon = 0) &= 0.2\,. \end{aligned}$$

You observe that the number of rabbit pairs in the second month is $x_2 = 2$, but you don't know how many rabbit pairs appeared on Laputa island in the first month x_1. You think that it must be one or two pairs, and therefore set a prior $P(x_1 = 1) = P(x_1 = 2) = 0.5$. What is the posterior probability $P(x_1 = 1|x_2 = 2)$ according to the model (M)?

Solution: $P(x_1 = 1|x_2 = 2) = \frac{1}{3} \approx 0.33$.
Indeed, using Bayes' rule, we get

$$\begin{aligned} P(x_1 = 1|x_2 = 2) &= \frac{P(x_2 = 2|x_1 = 1)P(x_1 = 1)}{P(x_2 = 2|x_1 = 1)P(x_1 = 1) + P(x_2 = 2|x_1 = 2)P(x_1 = 2)} \\ &= \frac{0.2 \cdot 0.5}{0.2 \cdot 0.5 + 0.4 \cdot 0.5} = \frac{1}{3}. \end{aligned}$$

(2 points) (ii) The ranger of the island tracked the number of rabbit pairs x_t during the 4 following months, i.e., for $t = 2, 3, 4, 5$. The data that he got is shown in Table 1.

x_2	x_3	x_4	x_5
2	5	11	25

Table 1: Observed rabbit paits at months $t = 2, 3, 4, 5$.

Based on this data the ranger asks you to reestimate the ω and provide the uncertainty level on your estimator, assuming that the noise parameter is $\epsilon \sim \mathcal{N}(0, \sigma_x^2 = 5)$. *(You can ignore the fact that this model cannot be exact due to the fact that $x_t \in \mathbb{N}$.)* You apply Bayesian linear regression with a Gaussian prior $\omega \sim \mathcal{N}(0, 1)$.

- What is the posterior mean of ω? (1 point)

Solution: $\mu_\omega = 2.2$.
Indeed, the X matrix is the vector $[2,\ 5,\ 11]$. The y is $[5,\ 11,\ 25]$. Then

$$X^T X = 2^2 + 5^2 + 11^2 = 4 + 25 + 121 = 150\ ,$$
$$X^T y = 2 \cdot 5 + 5 \cdot 11 + 11 \cdot 25 = 10 + 30 \cdot 11 = 340.$$

Hence, the posterior mean of ω is $\mu_\omega = (X^T X + \sigma_\epsilon^2 I)^{-1} X^T y = \frac{340}{150+5} = \frac{68}{31} \approx 2.2$.

- What is the posterior variance of ω? (1 point)

Solution: $\sigma_\omega^2 = \frac{1}{31}$.
Indeed, the variance is given by $\sigma_\omega^2 = (\sigma_x^{-2} X^T X + I)^{-1} = (\frac{150}{5} + 1)^{-1} = \frac{1}{31}$.

(2 points) (iii) After $T > 0$ observations, you estimate that the posterior distribution of ω is $\mathcal{N}(2, 0.035)$.

- Using model (M), and assuming again that $\epsilon \sim \mathcal{N}(0, 5)$ (centered Gaussian with variance 5), what is the variance σ_{T+1}^2 of x_{T+1} if $x_T = 20$? (1 point)

Solution: $\sigma_{T+1}^2 = 19$.
Indeed, the predicted variance is $\sigma_{T+1}^2 = 20^2 \cdot 0.035 + 5 = 14 + 5$, where 14 is the epistemic uncertainty part (which will decrease with more data), and 5 is the aleatoric part (which is intrinsic to the model).

- The variance σ_{T+1}^2 can be decomposed as $\sigma_{T+1}^2 = \sigma_e^2 + \sigma_a^2$, where σ_e^2 and σ_a^2 are the epistemic and the aleatoric variance respectively. What is the value of σ_e^2? (0.5 point)

Solution: $\sigma_e^2 = 14$.

- What is the value of σ_a^2? (0.5 point)

Solution: $\sigma_a^2 = 5$.

.5.5.5

(2.5 points) (iv) Now suppose you know that $\omega = 2$ and $\epsilon \sim \mathcal{N}(0, 5)$. And imagine that the ranger can only estimate the number of rabbit pairs up to some additive noise $\nu \sim \mathcal{N}(0, 2)$. Said differently, you get the following Kalman filter:

$$x_{t+1} = 2x_t + \epsilon \tag{1}$$
$$y_t = x_t + \nu\ . \tag{2}$$

1. Which of these sentences best describes this Kalman filter? (0.5 point)

◯ Equations (1) and (2) are the sensor and motion models, respectively.

√ **Equations** (1) **and** (2) **are the motion and sensor models, respectively.**

◯ Equations (1) and (2) define the motion model.

◯ Equations (1) and (2) define the sensor model.

2. Is the so-called *Kalman gain* used to update both the mean and the variance of the upcoming state? (0.5 point)

   √ **Yes.** ◯ No, just the mean. ◯ No, just the variance. ◯ None of them.

3. If one increased the variance of ν, would the Kalman gain increase? (0.5 point)

   ◯ Yes, increase. √ **No, decrease.** ◯ Stay the same. ◯ Cannot say.

4. If one increased the variance of ϵ, would the Kalman gain increase? (0.5 point)

   √ **Yes, increase.** ◯ No, decrease. ◯ Stay the same. ◯ Cannot say.

5. Can the Kalman gain be computed offline? (0.5 point)

   √ **Yes.** ◯ No. ◯ Cannot say.

2. Gaussian Processes (6 points)

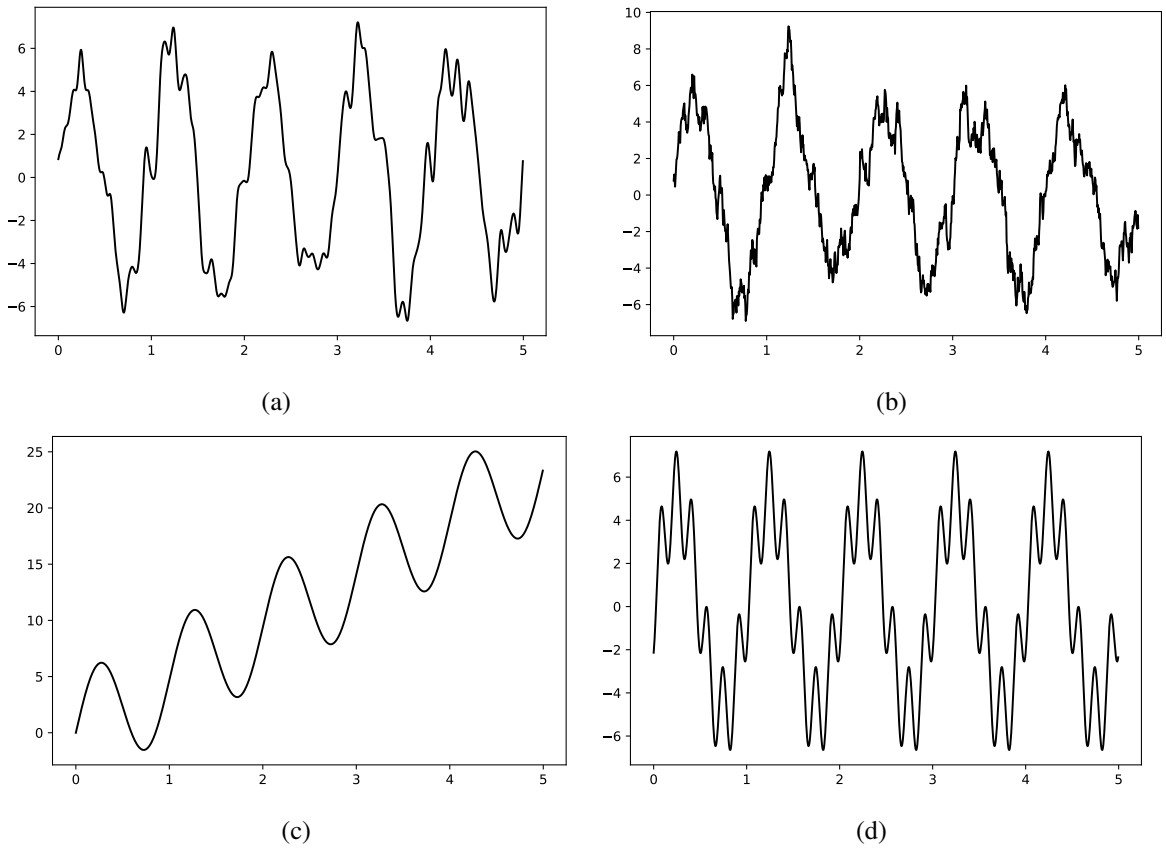


Figure 1: Five GP simulations with mean $\mu(x) = 5\sin(2\pi x)$, and using a different kernel for each plot.

Some company decides to model daily weather temperature fluctuations by a one-dimensional Gaussian process (GP) $\mathcal{GP}(\mu(\cdot), k(\cdot,\cdot))$, with covariance function k and mean function $\mu : x \mapsto 5\sin(2\pi x)$.

(2 points) (i) The company ran the 4 simulations shown in Figure 1): one for each of the following kernels (where $x, y \geq 0$ represent time stamps in some arbitrary units).

1. Gaussian kernel: $k(x, y) = \exp(-\lambda^2(x - y)^2)$.
2. Laplace kernel: $k(x, y) = \exp(-\lambda|x - y|)$.
3. Cosine kernel: $k(x, y) = \cos(2\lambda|x - y|)$.
4. Linear kernel: $k(x, y) = \lambda xy$.

The company used $\lambda = 6\pi$ and mean $\mu(x) = 5\sin(2\pi x)$ for all simulations. For all plots, the x- and y-axis represent time and temperature respectively in some arbitrary units. Could you help them by re-assigning each plot in Figure 1 to its most likely kernel?

- Which plot in Figure 1 corresponds to the Gaussian kernel? (0.5 point)
  ◯ (a) √ **(b)** ◯ (c) ◯ (d)

- Which plot in Figure 1 corresponds to the Laplace kernel? (0.5 point)
  ◯ (a) √ **(b)** ◯ (c) ◯ (d)
- Which plot in Figure 1 corresponds to the cosine kernel? (0.5 point)
  ◯ (a) ◯ (b) ◯ (c) √ **(d)**
- Which plot in Figure 1 corresponds to the linear kernel? (0.5 point)
  ◯ (a) ◯ (b) √ **(c)** ◯ (d)

Solution: The Laplace kernels yields rough paths, which gives (b). The cosine kernel must yield periodic variations around the mean μ, hence (d). The linear kernel can yield only linear functions, i.e. linear perturbations of the mean μ, hence (c). That leaves only (a) for the Gaussian kernel, which is confirmed by the smoothness of that curve.

(1 point) (ii) Now assume that you have only one noisy measurement of the temperature, $y_1 = f(x_1) + \epsilon_1$, where $\epsilon_1 \sim \mathcal{N}(0, 1)$ with a GP prior $\mathcal{GP}(\mu(\cdot), k(\cdot, \cdot))$ with mean $\mu(x) = \sin(2\pi x)$ and kernel $k(x, x') = \exp(-\|x - x'\|_2^2/2)$. What is the posterior updated mean for your model ?

◯ $\exp\left\{\frac{(x-x_1)^2}{2}\right\} y_1$

◯ $\exp\left\{-\frac{(x-x_1)^2}{2}\right\} (y_1 - \sin(2\pi x_1))$

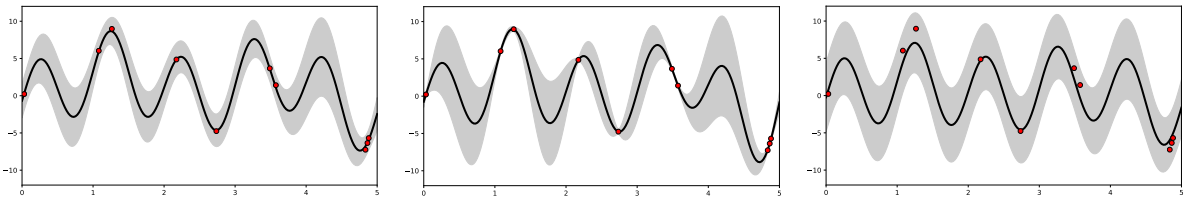
◯ $\sin(2\pi x) + \exp\left\{\frac{(x-x_1)^2}{2}\right\} \frac{1}{2}(y_1 - \sin(2\pi x_1))$

√ $\sin(2\pi x) + \exp\left\{-\frac{(x-x_1)^2}{2}\right\} \frac{1}{2}(y_1 - \sin(2\pi x_1))$

.5.5.5

(1.5 points) (iii) The company got a few additional measurements (x_i, y_i) shown as red dots in Figure 2. It assumed that the data could be modeled as $y_i = f(x_i) + \epsilon_i$ where f is a sample from a GP $\mathcal{GP}(\mu, k)$ with mean $\mu(x) = \sin(2\pi x)$ and Gaussian kernel $k(x, y) = \exp(-\lambda^2(x - y)^2)$ where $\lambda = 6\pi$, and where $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ is i.i.d. Gaussian noise. It then computed the posterior mean and variance functions for 3 different values of σ in $\{0.1, 0.8, 2.5\}$. The results are shown in Figure 2 (one plot for each value of σ). For all plots, the x- and y-axis represent time and temperature respectively in some arbitrary units.

- Which plot in Figure 2 corresponds to $\sigma = 0.1$? (0.5 point) ◯ (a) √ **(b)** ◯ (c)
- Which plot in Figure 2 corresponds to $\sigma = 0.8$? (0.5 point) √ **(a)** ◯ (b) ◯ (c)
- Which plot in Figure 2 corresponds to $\sigma = 2.5$? (0.5 point) ◯ (a) ◯ (b) √ **(c)**



(a) (b) (c)

Figure 2: Posterior mean (solid line) and variance (shaded) when modeling the data (red points) by a GP and using 3 different noise levels σ (one for each plot).

3. Approximation Methods

(2 points) (i) Which of the following statements about the Laplace approximation are true?

- The Laplace approximation is a good choice, if the target distribution is multi-modal. ◯ True √ **False**
- The Laplace approximation fits a normal distribution around the mode of the target distribution. √ **True** ◯ False
- The Laplace approximation requires knowledge of the normalizing constant of the target distribution. ◯ True √ **False**
- If the target distribution is a multivariate Gaussian distribution, the Laplace approximation recovers the target distribution exactly. √ **True** ◯ False

Solution: The Laplace approximation is not a good choice for multi-modal targets as it will tend underestimate the probability mass at the other modes significantly. If the underlying target distribution is Gaussian, the Laplace approximation recovers the mean and covariance of the target, and hence recovers the target distribution exactly. It does not require knowledge of the target distribution, as can be seen when deriving it using the second-order Taylor expansion.

(2 points) (ii) Consider a model $P(X, Z)$ for some observed data X and some unobserved data Z. We choose to approximate the posterior distribution $P(Z \mid X)$ by variational inference. Specifically, we choose $Q^* \in \mathcal{Q}$, such that

$$Q^* = \arg\min_{Q \in \mathcal{Q}} \mathrm{KL}(Q(Z) \parallel P(Z \mid X))$$

Which of the following choices define an equivalent objective?

- $Q^* = \arg\min_{Q \in \mathcal{Q}} \mathrm{KL}(P(Z|X) \parallel Q(Z))$ ◯ True √ **False**
- $Q^* = \arg\max_{Q \in \mathcal{Q}} \mathbb{E}_{Z \sim Q(Z)} [\log(P(X, Z)]$ ◯ True √ **False**
- $Q^* = \arg\min_{Q \in \mathcal{Q}} \mathbb{E}_{Z \sim Q(Z)} [\log(Q(Z))] + \mathrm{KL}(Q(Z) \parallel P(Z))$ ◯ True √ **False**
- $Q^* = \arg\max_{Q \in \mathcal{Q}} \mathbb{E}_{Z \sim Q(Z)} [\log(P(X \mid Z)] - \mathrm{KL}(Q(Z) \parallel P(Z))$ √ **True** ◯ False

Solution: The KL divergence is not symmetric. The second option is missing the entropy term for Q. The third option is wrong. The fourth option is correct.

(2 points) (iii) Which of the following statements about the reparameterization trick are true?

- It requires that the random variable is continuous. $\surd$ **True** ◯ False
- It reduces variance by introducing bias. ◯ True $\surd$ **False**
- It cannot be used to optimize Monte Carlo objectives, e.g., $\mathbb{E}_{X\sim p(\cdot|\lambda)}\left[f(X)\right]$, with respect to distributional parameters, i.e., λ. ◯ True $\surd$ **False**
- It can often be implemented with ease using software for automatic differentiation. $\surd$ **True** ◯ False

Solution: The reparameterization trick requires that the random variable is continuous, because the function g (Lecture 5, Slide 30) needs to be invertible. It gives an unbiased gradient estimator for the continuous distribution it reparametrizes, hence it does not introduce bias. It works for MCO of the given form, that is the point. It works well with auto-diff software, which is part of the reason why it is very popular.

4. Sampling methods

(2 points) (i) Which of the following statements about Markov Chain Monte Carlo (MCMC) methods are true?

- MCMC methods can only be used to sample from log-concave distributions.
  ◯ True √ **False**
- If $\{X_1, X_2, \ldots\}$ is an ergodic Markov chain with stationary distribution π and f is a bounded function $[0, C]$, it holds that $\mathbb{P}\left(\left|\mathbb{E}_{X\sim\pi}\left[f(X)\right] - \frac{1}{N}\sum_{i=1}^{N} f(X_i)\right| > \epsilon\right) \leq 2e^{\frac{-2N\epsilon^2}{C}}$.
  ◯ True √ **False**
- If the transition probabilities are symmetric, such that $P\left(X' \mid X\right) = P\left(X \mid X'\right)$, the detailed balance equation is satisfied for the uniform distribution. √ **True** ◯ False
- The stochastic gradient Langevin dynamics (SGLD) algorithm is guaranteed to converge to the target distribution for a constant step size. ◯ True √ **False**

Solution: The MCMC samples are positively correlated, the variance typically does not decrease at the same rate as when the samples are independent. MCMC methods can be used to sample from any distribution, not just log-concave distributions. The statement about the detailed balance equation is indeed correct, which follows directly from its definition and we do require a "burn-in" phase for SGLD.

(2 points) (ii) Consider a Markov chain in discrete times $X_1, X_2, \ldots$, where the variable X_i may assume either of two states, i.e., $X_i \in \mathcal{S} = \{A, B\}$. The transition model of the Markov chain is illustrated in Figure 3. This Markov chain has a unique stationary distribution $\lim_{N\to\infty} \mathbb{P}\left(X_n = x\right) = \pi(x)$ for $x \in \mathcal{S}$. What is $\pi(A)$?

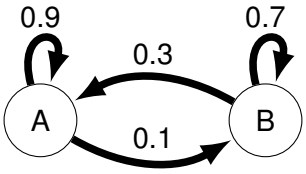


Figure 3: Graphical illustration for the transition probabilities of the Markov model. For example, the probability associated with a transition from state A to state B from time step t to time step $(t+1)$ is 0.1.

Solution: The transition matrix is $P = \begin{bmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{bmatrix}$ and to find the stationary distribution π, we solve the system $\pi(P - I) = 0$, which gives $\pi = (0.75, 0.25)$. Hence, $\pi(A) = 0.75$.

(2 points) (iii) For educational purposes, consider sampling from a bivariate normal distribution using various Markov Chain Monte Carlo (MCMC) schemes. In particular, consider the following bivariate normal distribution as your target,

$$X \sim P = \mathcal{N}(\mu, \Sigma) \quad \text{with } \mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ and } \Sigma = \begin{bmatrix} 1 & 0.9 \\ 0.9 & 1 \end{bmatrix},$$

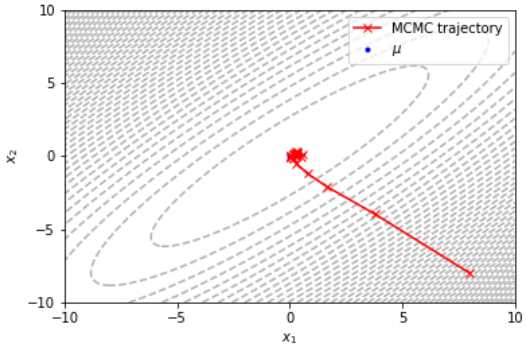
and the following four MCMC schemes:

1. Random walk Metropolis Hastings (with small variance):
   $R(x' \mid x) \sim \mathcal{N}(x' \mid x, \sigma I)$ with $\sigma = 0.1$
2. Random walk Metropolis Hastings (with large variance):
   $R(x' \mid x) \sim \mathcal{N}(x' \mid x, \sigma I)$ with $\sigma = 1.0$
3. Gibbs sampler alternating between dimensions, in particular
   $R(x_1'|x_2) = \mathcal{N}\left(x_1' \mid \overline{\mu}_1, \overline{\Sigma}_1\right)$ and $R(x_2'|x_1) = \mathcal{N}\left(x_2' \mid \overline{\mu}_2, \overline{\Sigma}_2\right)$, where $\overline{\mu}_1, \overline{\mu}_2, \overline{\Sigma}_1, \overline{\Sigma}_2$ depend on x_1, x_2, μ and Σ respectively.
4. Metropolis adjusted Langevin algorithm:
   $R(x' \mid x) \sim \mathcal{N}(x' \mid x + \tau \nabla \log \pi(x), 2\tau I)$ with $\tau = 0.1$, where π is the density of P.

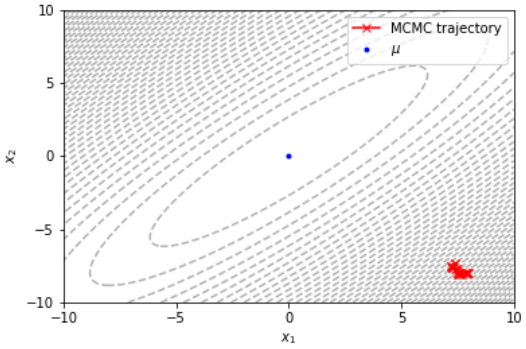
You choose $X_0 = [8, -8]$ as your initial state and run each MCMC scheme for 25 iterations to generate a trajectory over states $\mathcal{T}_i = \{X_0^i, X_1^i, \ldots X_{25}^i\}$ for $i \in \{1, 2, 3, 4\}$. The four trajectories are visualized in Figure 4 (in no particular order).

- Which trajectory in Figure 4 corresponds to sampler 1 (random walk with small variance)? (0.5 point)
  ◯ Trajectory A √ **Trajectory B** ◯ Trajectory C ◯ Trajectory D
- Which trajectory in Figure 4 corresponds to sampler 2 (random walk with large variance)? (0.5 point)
  ◯ Trajectory A ◯ Trajectory B ◯ Trajectory C √ **Trajectory D**
- Which trajectory in Figure 4 corresponds to sampler 3 (Gibbs sampler)? (0.5 point)
  ◯ Trajectory A ◯ Trajectory B √ **Trajectory C** ◯ Trajectory D
- Which trajectory in Figure 4 corresponds to sampler 4 (Metropolis adjusted Langevin)? (0.5 point)
  √ **Trajectory A** ◯ Trajectory B ◯ Trajectory C ◯ Trajectory D

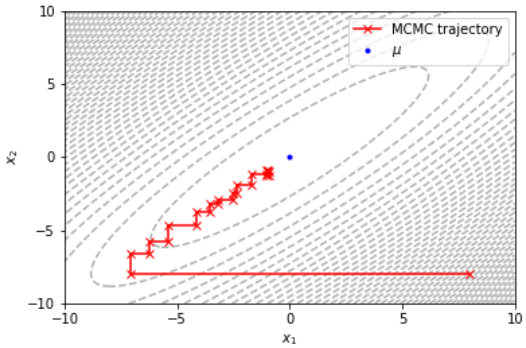
Solution: The solution is 1B, 2D, 3C, 4A. Trajectory A corresponds to MALA, the trajectory moves very clearly towards the mean (ascent on the density) and then explores the neighborhood close to the mean. Trajectory B only explores narrowly in the neighborhood close to the initial state, typical of RW where the variance is chosen too small. Trajectory C corresponds to the Gibbs sampler as can be clearly seen from the stair-like trajectory for the visited states. Trajectory D corresponds to RW with large variance. There is slow progress towards the mean, there are also fewer visited states as a result of a larger number of rejected proposals.



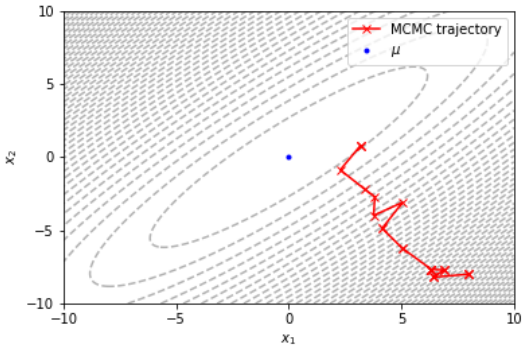
Trajectory A



Trajectory B



Trajectory C



Trajectory D

Figure 4: Contour plots for the target distribution P and trajectories for the four MCMC schemes.

5. Bayesian optimization (9 points)

Let f be a function sampled from a Gaussian Process (GP) with mean function 0 (i.e., the null function) and Gaussian covariance function, and consider the sequential procedure where at every round t, we select x_t from a given domain D and receive a noisy observation:

$$y_t = f(x_t) + \epsilon_t, \quad \epsilon_t \overset{iid}{\sim} \mathcal{N}(0, \sigma^2).$$

To select points at every round t, we consider the uncertainty sampling rule

$$x_t = \operatorname*{argmax}_{x \in D} \sigma^2_{t-1}(x), \tag{3}$$

where $\sigma^2_{t-1}(\cdot)$ denotes GP posterior variance. Further, let x^* a maximizer of f over the domain D, i.e., $x^* \in \operatorname{argmax}_{x \in D} f(x)$.

(3 points) (i) Which of the following statements about *uncertainty sampling* are correct?

- Uncertainty sampling is equivalent to greedily maximizing information gain in the considered homoscedastic noise case. √ **True** ◯ False
- Uncertainty sampling is equivalent to greedily maximizing information gain in case of heteroscedastic noise. ◯ True √ **False**
- Uncertainty sampling is suitable for exploration and active learning. It does not explicitly optimize regret. √ **True** ◯ False
- As $t \to \infty$, it holds that $f(x_t) \to f(x^*)$. ◯ True √ **False**
- As $t \to \infty$, it holds that for $\hat{x}_t = \operatorname{argmax}_{x \in D} \mu_t(x)$ we have $f(\hat{x}_t) \to f(x^*)$, where $\mu_t(\cdot)$ is the posterior mean of the GP. √ **True** ◯ False
- Let $\boldsymbol{x}_{1:T}$ be the vector of inputs $\{x_1, \ldots, x_T\}$ selected via (3), and $\boldsymbol{y}_{1:T}$ be the corresponding vector of noisy observations. Let $I(\cdot\,;\cdot)$ denote mutual information and let $\gamma_T := \max_{S \subset \{1,\ldots,T\}, |S| \leq T} I(f(\boldsymbol{x}_S); \boldsymbol{y}_S)$ be the so-called maximum information gain. Then it holds that $\gamma_T \leq (1 - 1/e)^{-1} I(f(\boldsymbol{x}_{1:T}); \boldsymbol{y}_{1:T})$. √ **True** ◯ False

(1 point) (ii) Instead of maximizing over the whole domain D in (3), one can keep the set of *plausible maximizers* M_t, and select $x_t = \operatorname{argmax}_{x \in M_t} \sigma^2_{t-1}(x)$ from M_t instead.

Let $\mathrm{ucb}_t(x) := \mu_{t-1}(x) + \beta_t \sigma_{t-1}(x)$ and $\mathrm{lcb}_t(x) := \mu_{t-1}(x) - \beta_t \sigma_{t-1}(x)$ be the upper and lower confidence bounds and β_t be such that $f(x) \in [\mathrm{lcb}_t(x), \mathrm{ucb}_t(x)]$ for every x, t. Which of these sets could be best used as a set of plausible maximizers? (Said differently: pick the smallest set M_t of inputs that, based on the upper and lower confidence bounds, contains with high probability the set of all maximizers of the function f.)

√ $M_t = \{x \in D : \mathrm{ucb}_{t-1}(x) \geq \max_{x \in D} \mathrm{lcb}_{t-1}(x)\}$

◯ $M_t = \{x \in D : \mathrm{ucb}_{t-1}(x) \leq \mu_{t-1}(x)\}$

◯ $M_t = \{x \in D : \mu_{t-1}(x) \geq \mathrm{lcb}_{t-1}(x)\}$

◯ $M_t = \{x \in D : \mathrm{ucb}_{t-1}(x) \leq \max_{x \in D} \mathrm{lcb}_{t-1}(x)\}$

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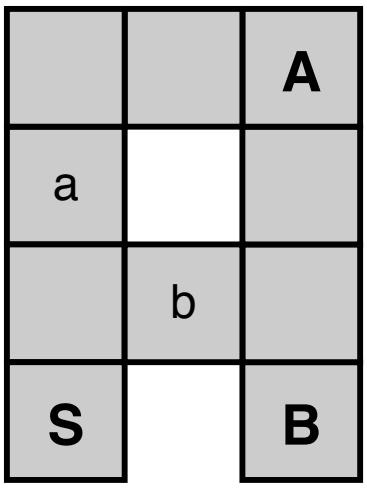
(2.5 points) (iii) It can be shown that the modified uncertainty sampling from the previous question, i.e., $x_t = \text{argmax}_{x \in M_t} \sigma^2_{t-1}(x)$, can achieve sublinear regret $R_T = \sum_{t=1}^{T} (\max_{x \in D} f(x) - f(x_t))$.

Which of the following statements regarding *regret* are correct?

- By definition, the regret is sublinear if $R_T/T \to c$ for some constant $c > 0$, as $T \to \infty$.
  ◯ True √ **False**
- Let $r_T = \max_{x \in D} f(x) - f(x_T)$. Vanishing instantaneous regret $r_T \to 0$ implies sublinear regret. √ **True** ◯ False
- Establishing sublinear regret bounds for GP-UCB (UCB = upper confidence bound) is possible in case of squared-exponential but not in case of linear kernel. ◯ True √ **False**
- At every round, both GP-UCB and modified uncertainty sampling trade-off exploration and exploitation and only sample points from the current set of plausible maximizers.
  √ **True** ◯ False
- Sublinear regret implies $\max_t f(x_t) \to f(x^*)$. √ **True** ◯ False

6. A-Maze (9 points)

Consider the grid world shown below. The agent starts in state S. In each round, the agent chooses actions to go to a neighbouring state or stay in the same state. All transitions are deterministic. If the agent is in state A or B at the beginning of a round, it receives reward $r_A > 0$ and $r_B > 0$ respectively and the episode ends (i.e. A and B are terminal states). The reward in all other states is zero. We use a discount-factor $0 < \gamma < 1$.



(1 point) (i) Assume that the reward in A is $r_A = +3$ and the reward in B is $r_B = +1$, and the discount factor is $\gamma = 0.5$. Which of the following polices are optimal for the given MDP?

◯ Going through a directly to A is optimal, and all other policies are suboptimal.
◯ Going through b directly to A is optimal, and all other policies are suboptimal.
√ **Going through either a or b directly to A is optimal, and all other policies are suboptimal.**
◯ Going through b directly to B is optimal, and all other policies are suboptimal.

Solution: Going through either a or b directly to A is optimal, and all other policies are suboptimal. Going to A is optimal and this can be done either through a or b as the number of steps is the same. Going to B is suboptimal as: $r_A * \gamma > r_B$.

(1 point) (ii) For what values of $r_A > 0, r_B > 0$ and $0 < \gamma < 1$ is the optimal policy *unique*?

√ $\gamma < r_B/r_A$
◯ $\gamma \leq r_B/r_A$

◯ $\gamma = r_B/r_A$
◯ $\gamma \geq r_B/r_A$
◯ $\gamma > r_B/r_A$

Solution: Correct solution is $\gamma < r_B/r_A$. First, we observe that whenever the optimal policy ends up in A it is not unique (see the previous question), hence this rules out the case $\gamma > r_B/r_A$. From the previous question, we also have that the optimal policy ends up in B if $r_A * \gamma < r_B$. This rules out all the answers that involve $=, \geq, \leq$, together with the condition $\gamma \in (0, 1)$.

(3 points) (iii) A way to compute the optimal policy for the given MDP is *policy iteration*. Which of the following statements about policy iteration is correct?

- In each iteration, *policy iteration* computes the value for the *current* policy. √ **True** ◯ False
- In each iteration, the value of the *current* policy can be computed efficiently by solving a single linear system of equations. √ **True** ◯ False
- In each iteration, we need to solve the *Bellman optimality equation*, which is a non-linear system of equations. ◯ True √ **False**
- In each iteration, the value of the *current* policy can be computed using *value iteration*. ◯ True √ **False**
- Policy iteration converges in a finite number of iterations. √ **True** ◯ False
- There exist a discount factor $0 < \gamma < 1$ and reward values r_A and r_B, for which policy iteration never converges, but the approximation error in ℓ_∞-norm of the value function goes to zero. ◯ True √ **False**

Solution: Correct solutions are: True, True, False, True, False, True, False. These are textbook questions and answers.

(1 point) (iv) Assume that our initial policy for *policy iteration* is defined to stay in the current state (in all states). Consider again the values for r_A, r_B and γ used in the first question, i.e., $r_A = 3$, $r_B = 1$ and $\gamma = 0.5$.

How many policy updates do we need at most to obtain the optimal policy?

Solution: The correct solution is 4. In the first iteration, the policy is adjusted at the neighbouring state of A and B, then in the second iteration, at the neighbouring states of the previous neighbours, and so on.

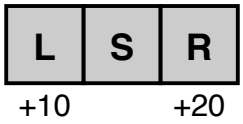
7. Reinforcement Learning (9 points)

(2 points) (i) Are the following methods for reinforcement learning *model-based* or *model-free*? (Hint: model-based means that the algorithm explicitly estimates the transition probabilities of the MDP)

- Q-learning with tabular representation: ◯ model-based √ **model-free**
- Q-learning with function approximation: ◯ model-based √ **model-free**
- Policy search: ◯ model-based √ **model-free**
- Rmax: √ **model-based** ◯ model-free

> **Solution:** Q-learning estimates the Q-function without estimating the transition kernel, hence is model-free. Policy search directly parametrizes the policy, hence also model-free. Rmax estimates the transition kernel, hence is model-based.

For the remaining questions we consider the three-state MDP shown below. The agent starts in state S.



In each state the agent has two actions "left" and "right", $\{\rightarrow, \leftarrow\}$, which deterministically move the agent to the corresponding neighbouring state. When this is not possible, i.e. if $\leftarrow$ is chosen in L or $\rightarrow$ is chosen in R the agent stays in the same state. The only non-zero rewards are obtained when transitioning to L or R. Specifically:

- $r(S, \leftarrow) = r(L, \leftarrow) = +10$,
- $r(S, \rightarrow) = r(R, \rightarrow) = +20$.

(3 points) (ii) A behaviour policy produces the following observations:

1. $(S, \rightarrow, +20, R)$,
2. $(R, \rightarrow, +20, R)$,
3. $(R, \leftarrow, +0, S)$,
4. $(S, \leftarrow, +10, L)$.

We initialize the value function with $V_0(s) = 0$ for all $s \in \{L, S, R\}$. Compute the values $V_4(s)$ of the value function obtained by TD-learning with learning rate $\alpha = 0.5$ and discount factor $\gamma = 1$ for the observations in the following order.

- What is the value of $V_4(L)$? (1 point)

> **Solution:** $V_4(L) = 0$. (Justification below.)

- What is the value of $V_4(S)$? (1 point)

  Solution: $V_4(S) = 10$ (Justification below.)

- What is the value of $V_4(R)$? (1 point)

  Solution: $V_4(R) = 10$. (Justification below.)

Solution: The updates are the following:

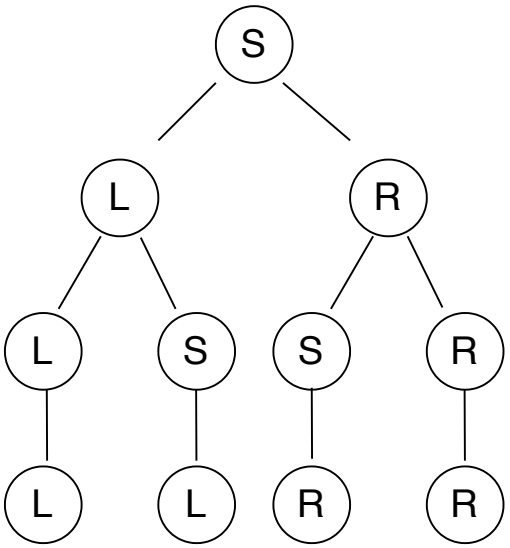
1. $(S, \rightarrow, +20, R)$, $V_1(S) = 0.5V_0(S) + 0.5(20 + V_0(R)) = 10$
2. $(R, \rightarrow, +20, R)$, $V_2(R) = 0.5V_1(R) + 0.5(20 + V_1(R)) = 10$
3. $(R, \leftarrow, +0, S)$, $V_3(R) = 0.5V_2(R) + 0.5(0 + V_2(S) = 10$
4. $(S, \leftarrow, +10, L)$, $V_4(S) = 0.5V_3(S) + 0.5(10 + V_3(L)) = 10$

Hence $V_4(L) = 0$, $V_4(S) = V_4(R) = 10$.

(3 points) (iii) For the same MDP, apply Q-learning, where the behaviour policy is set to the greedy policy according to the current estimated Q-values: $a_t = \arg\max_{a \in \{\leftarrow, \rightarrow\}} Q_{t-1}(s_t, a)$. When there is ambiguity, we uniformly randomize between the two actions.

We initialize the Q-function with $Q_0(s, a) = 0$ for all $s \in \{L, S, R\}$ and $a \in \{\leftarrow, \rightarrow\}$. Use learning rate $\alpha = 0.5$ and discount factor $\gamma = 1$. What is the probability p that the agent is in state R after 4 steps?

Solution: $p = 0.5$.
Indeed, the transition tree is given below. When a node has two children, each path is chosen with probability 0.5.



Details:

- In the first step, we are in S. We have $Q_0 \equiv 0$, hence $\{\leftarrow, \rightarrow\}$ is chosen uniformly.
- Let's say we transition to R (the argument is symmetric if we go to L), then we observe $(S, \rightarrow, +20, R)$, and we update $Q_1(S, \rightarrow) = 10$.
- In state R, greedy is inconclusive, so we randomize again.
  - If we pick $\rightarrow$, the update is $(R, \rightarrow, +20, R)$, so we have $Q_2(R, \rightarrow) = 10$. Hence afterwards the greedy action is $\rightarrow$ again.
  - If we pick $\leftarrow$, the update is $(R, \leftarrow, +0, S)$, so we have $Q_2(R, \leftarrow) = 5$. Hence afterwards the greedy action is $\rightarrow$, and we go back to R.

Hence the probability to end up in the R state is 0.5 (and this will not change in further steps).